

Advanced Data Analytics — Lecture 1b

Crash-course on the Computational Infrastructure

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Slides by Simon Scheidegger (2023–2025), adapted with permission

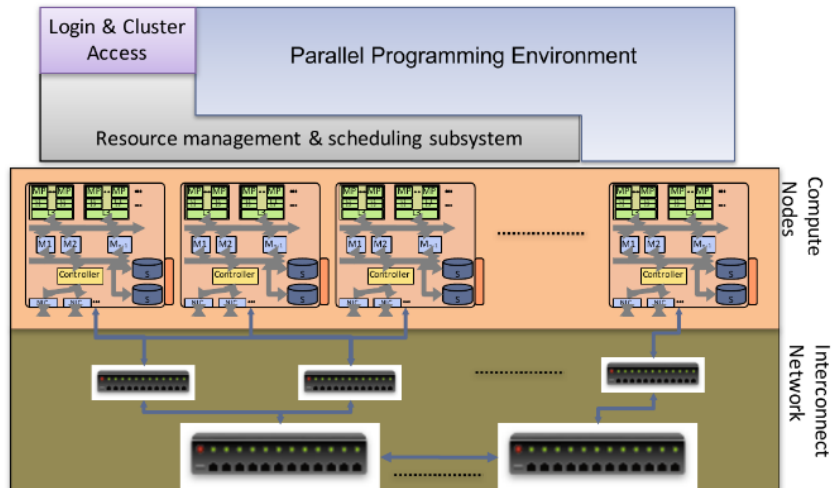
September 14th, 2026 | 12:30 - 14:00: Internef 125 | 16:30 - 18:00: Anthropole 3185

- ▶ Crash-course on the Computational Infrastructure
 - ▶ Make first steps on a Linux Cluster (Login, short overview of basic unix commands like cd, pwd, cp, scp,...)
 - ▶ Get lecture notes/codes for the lecture
- ▶ Hello World in Python

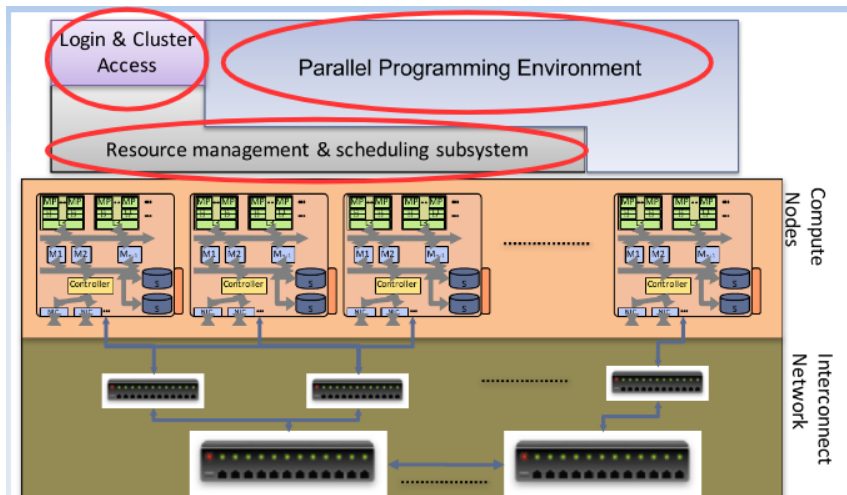
**MORE DETAILS IN THE TA
SESSION**

- ▶ For this course, we use Nuolos.cloud: JupyterLab in the browser, every package installed, nothing on your laptop.
- ▶ Its setup is very similar to any other system
- ▶ Enrol *once* through the enrolment link on Moodle, with the email you use for Moodle; afterwards log in at <https://app.nuolos.cloud>.
- ▶ Step-by-step guide: <https://unil-ada.github.io/nuolos.html>

An abstract compute cluster



An abstract compute cluster



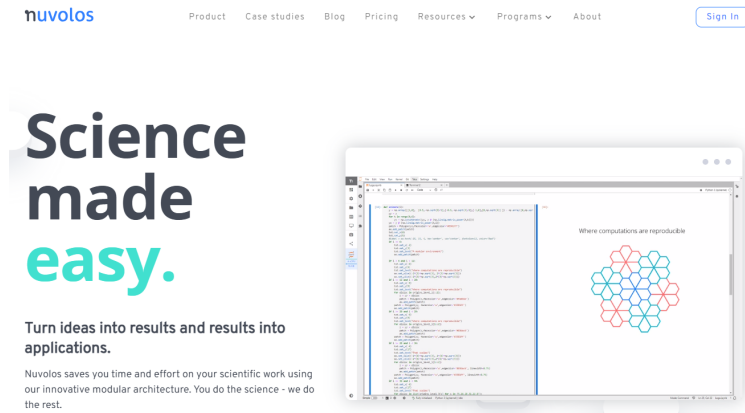
A real compute cluster



Credit: Blue Waters research cluster at NCSA, Illinois, USA (active 2007-2021).

Access to Nuvolos Services

ENROL NOW: app.eu1.nuvolos.cloud/enroll/class/EcQWpcx0Kr4
then log in, every time, at <https://app.nuvolos.cloud>



The image shows the Nuvolos website header and main content area. The header includes the Nuvolos logo, navigation links for Product, Case studies, Blog, Pricing, Resources, Programs, and About, and a Sign in button. The main content area features the text "Science made easy." and a subheading "Turn ideas into results and results into applications." Below this, a paragraph states: "Nuvolos saves you time and effort on your scientific work using our innovative modular architecture. You do the science - we do the rest." To the right, there is a screenshot of a web application interface. The interface has a code editor on the left with Python code and a right panel with the text "Where computations are reproducible" and a hexagonal molecular structure graphic.

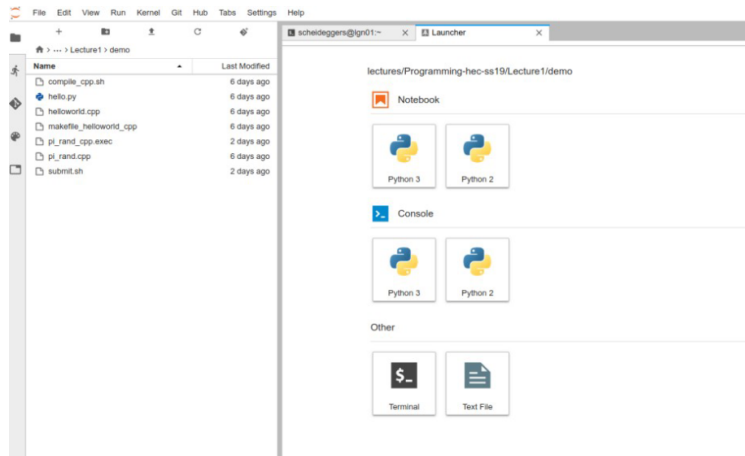
Science made easy.

Turn ideas into results and results into applications.

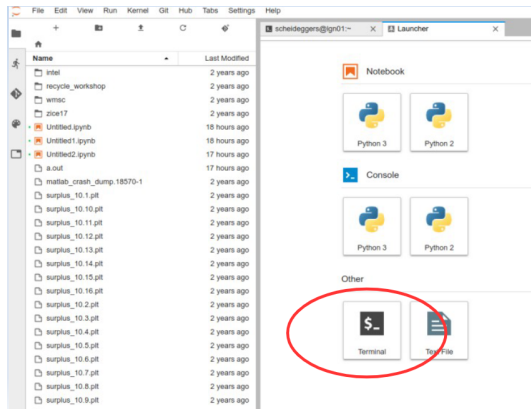
Nuvolos saves you time and effort on your scientific work using our innovative modular architecture. You do the science - we do the rest.

Where computations are reproducible

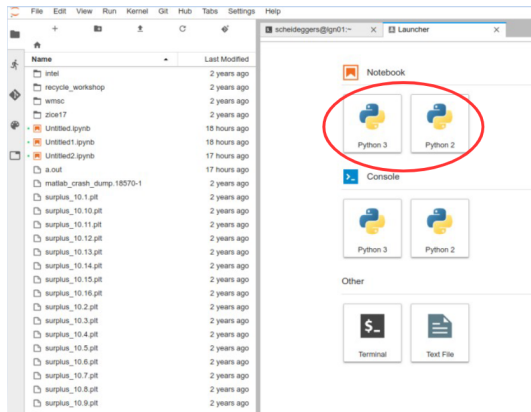
Login via Browser



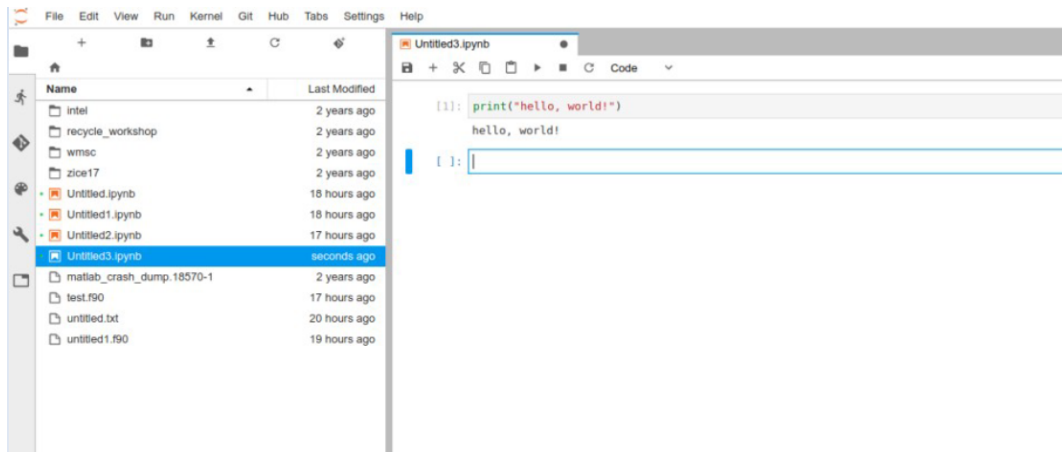
Login via Browser



Login via Browser



Login via Browser — Jupyter



Basic Linux commands

Command	Description
<code>pwd</code>	Print name of current/working directory
<code>cd [Directory]</code>	Change directory (no directory → change to home)
<code>ls [Directory]</code>	List directory contents (no directory → list current)
<code>cat FILE</code>	Concatenate files and print on the standard output
<code>mkdir DIRECTORY</code>	Make directories
<code>mkdir -p DIRECTORY</code>	Make directories, make parent directories as needed
<code>cp SOURCE. . . DIRECTORY</code>	Copy files and directories
<code>cp -r SOURCE. . . DIRECTORY</code>	Copy files and directories, copy directories recursively
<code>mv SOURCE. . . DIRECTORY</code>	Move (rename) files
<code>man COMMAND</code>	An interface to the on-line reference manuals

Basic Linux commands

Command	Description
<code>ssh -X foo@host.com</code>	OpenSSH SSH client (remote login program), access to host.com with user foo
<code>scp foo@host.com:/home/bar ./</code>	Secure copy (remote file copy program), copy file bar from /home on host.com to directory
<code>scp bar foo@host.com:/home/</code>	Secure copy (remote file copy program), copy file bar from the local host to /home on host.com
<code>git clone git@github.com:whatever folder-name</code>	The stupid content tracker, Clone a repository (whatever) into a new directory (folder-name)
<code>git checkout</code>	Checkout a branch or paths to the working tree.